End term Exam

CCC - Algorithmic thinking.

Q1: let A be a two Dimensional array such that
(5)

$$A[lb_1 \ldots ub_1 , lb_2 \ldots ub_2]$$

where lb_1 and ub_1 is the lower and upper index of row and lb_2 and ub_2 is the lower and upper index of column.

let base address is L_0 and C is the size of element then what is the Location of

$Loc(A[i][j])$ Assuming row major memory.

Q2 Arrange these asymptotic functions in increasing order of growth.
(2)

$$n, n^3, 2^n, n!, \log n, n \log n, n^2$$

Q3 How many algorithm can you write for Solving to find the prime numbers? Indicate which is the simplest and the most efficient?
(5)

Q4 A Sender is employing public key cryptography to
(2) send a secret message to a receiver. Which one of the following statement is true ?

A). Sender encrypts using receivers public key.
B) Sender encrypts using his own public key.
C) Receiver decrypts using senders public key.
D) Receiver decrypts using his own public key

Q5 The total no. of keys required for a set of
(4) n individuals to be able to communicate with each other using secret key and public key cryptosystems respectively are _____ & _____
Explain.

Q6 using RSA suppose $P=11$ & $q=3$, and $e=3$
(6) find d and encrypt the $M=14$.
Show the process of encryption & Decryption

Q7 what is hamming distance ?
(2)

Q

Q8 There are 20 priests in a temple. One day
(6) Lord Ganesha appears before them and tells them that some of them have sinned, and that a black spot would appear on the forehead of all the priests who have sinned. The priests are not allowed to look into a mirror or communicate with each other. When any priest finds out that there is a spot on his forehead, he would leave the temple on that day itself. At least 1 priest has sinned. How can a priest find out whether he has a spot on his forehead. What would be the pattern of the priests leaving the temple?

Q9. Pages of a book are numbered sequentially
(4) starting with 1. If the total number of decimal digits used is equal to 1578, how many pages are there in the book?

Q10. There are n people numbered 1 to n standing
(4) in a circle. Starting the count with person 1 every second person is eliminated until only one person is left. Where in the circle should a person stand to remain the last person standing